ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
07-05-2021

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE [SCORE-II]

PAPER-1

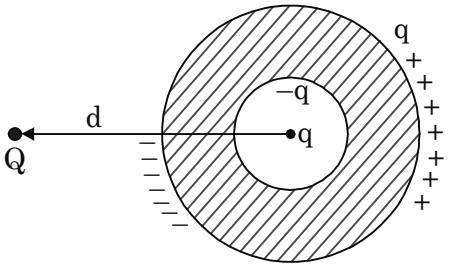
PART-1 : PHYSICS — SOLUTION

SECTION-I

1. Ans. (A or D)

Sol. Potential due to sphere at centre

$$= kq\left[\frac{1}{b} - \frac{1}{a}\right]$$



$$\text{Potential energy } = -\frac{k^2q}{a} + \frac{kq^2}{b}$$

2. Ans. (B)

Sol. $Ai = \lambda V_i$ and $A_f = \lambda N_f$

No. of α-particles emitted = $N_i - N_f$

$$= \frac{1}{\lambda}[A_i - A_f] = \frac{T}{\ell n 2}[A_i - A_f]$$

3. Ans. (B)

Sol. Equivalent E.M.F. $= \dfrac{\frac{18}{6} + \frac{3}{3}}{\frac{1}{6} + \frac{1}{3}} = 8V$

Final charge stored on capacitor = $16 \times 8\ \mu C = 128\ \mu F$

Charge flower through battery = $(128 - 64)\mu C = 64\mu C$

Work done by battery $w = 8 \times 64 = 512\ \mu J$

4. Ans. (A)

Sol. At node, energy is maximum when all particle reach to there extreme position.

5. Ans. (A)

Sol. Light rays after passing from lens must be parallel to principal axis to retracte path after reflection from mirror.

$$\frac{1}{45} = \left[\frac{3}{2} - 1\right]\left[\frac{1}{R} - \frac{1}{(-R)}\right]$$

R = 45 cm

6. Ans. (D)

Sol. Remove 10Ω resistance as wheatstone is balanced.

7. Ans. (B)

Sol. The wavelength at which the intensity of radiation is maximum depends only on the temperature. As this wavelength is the same for both, their temperatures must be equal. The difference in intensities can then be due only to a difference in their emissivities.

8. Ans. (D)

Sol. Applying angular momentum conservation

$I_0\omega_0 = I\omega$

$$\omega = \frac{I_0\omega_0}{I}$$

Angular momentum of small rings

$$= \frac{2mx^2 I_0\omega_0}{\left(\frac{MR^2}{2} + 2mx^2\right)}$$

$x \rightarrow$ perpendicular distance of small rings from the axis.

x first increases and then decreases.

9. Ans. (A,C)

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10. Ans. (B,D)

Sol. The fan is running at 200 V, consuming 1000 W, then $I = \frac{1000}{200} = 5A$

But as coil resistance is 1Ω, power dissipated by internal resistance as heat is $P1 = I^2R = 25W$.

If V is the net e.m.f. across the coil, then

$$\frac{V^2}{R} = 25W \quad \text{or} \quad V = 5V$$

Net e.m.f. = source e.m.f. – back e.m.f.

or $V = VS - e \Rightarrow e = 195$ V

11. Ans. (A,C)

Sol. $g = \frac{GM}{R^2} = \frac{6.6742 \times 10^{-11} \times 6 \times 10^{24}}{6.4^2 \times 10^{12}} = 9.77666$

$$\frac{dg}{g} = \frac{dG}{G} + \frac{dM}{M} + \frac{2dR}{R}$$

$$= \frac{0.001}{6.6742} + 0.0001 + 2 \times 0.0002 = 0.00065$$

$\therefore \quad g = (9.77767 \pm 0.0007)\ m/s^2$

5 significant figure

12. Ans. (B)

Sol. $v = \sqrt{\frac{GM}{R}}$

$$\frac{dv}{v} = \frac{1}{2}\frac{dM}{M} + \frac{1}{2}\frac{dR}{R} + \frac{1}{2}\frac{dG}{G}$$

$$= \frac{1}{2}\left[0.01 + 0.02 + \frac{0.001}{6.6742} \times 100\right]\%$$

$= 0.0225\%$

SECTION-II

1. Ans. 2.00

Sol. $$\frac{\int \vec{B} \cdot d\vec{\ell}\ \text{CDA}}{\int \vec{B} \cdot d\vec{\ell}\ \text{ABC}} = \frac{\frac{\mu_0 I}{2} + \frac{\mu_0 I}{2\pi}\frac{\pi}{6} \times 2}{\frac{\mu_0 I}{2} - \frac{\mu_0 I}{2\pi}\left(\frac{\pi}{6}\right) \times 2} = 2$$

2. Ans. 0.20

Sol. $ma = mg - F_b$

Density of ball of larger radius $\rho_1 = 2\rho$

Density of ball of larger radius $\rho_2 = \frac{4\rho}{3}$

After a long time, ball will acquire terminal speed

$$V = \frac{2}{9}R^2 g \frac{(\rho - \rho_L)}{\eta}$$

3. Ans. 2.25

Sol. $W = 2\omega_0 = \frac{2}{\sqrt{LC}}$ and liner $XL = WL = 2\sqrt{\frac{L}{C}}$;

$$X_C = \frac{1}{WC} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

$P = VI \cos\phi$

$$P = \frac{V^2 R}{R^2 + (X_L - X_C)^2}$$

Energy supplied $E = PT = \frac{\pi R\sqrt{LC}V^2}{R^2 + 2.25\left(\frac{L}{C}\right)}$

4. Ans. 0.75

Sol. $T = 2\pi\sqrt{\frac{I}{Mgd}} = 2\pi\sqrt{\frac{\frac{3}{2}MR^2}{MgR}}$

$$\Rightarrow \quad 2\pi\sqrt{\frac{3R}{2g}} = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\ell = \frac{3}{2}R$$

5. Ans. 4.90

Sol. $y = \frac{mg}{A}\frac{\ell}{\Delta\ell} = \frac{Mg}{\pi r^2}\frac{\ell}{\Delta\ell}$

$$\frac{\Delta y}{y} \times 100 = \left[\frac{2\Delta r}{r} + \frac{\Delta\ell}{\ell} + \left[\Delta\frac{(\Delta\ell)}{\Delta\ell}\right]\right] \times 100$$

$$= \left[2 \times \frac{0.001}{0.05} + \frac{0.1}{100} + \frac{0.001}{0.125}\right] \times 100$$

$$= \left[\frac{2}{50} + \frac{1}{1000} + \frac{1}{125}\right] \times 100$$

$= [0.04 + 0.001 + 0.008] \times 100 = 4.90$

6. Ans. 8.00

Sol. Change in momentum of 1 particle = $2Mv_0\cos\theta$

No. of particle striking per second per unit length

$= (nv)\ Rd\theta\ \cos\theta$

$$f = \int_{-\pi/2}^{\pi/2} (nvR\cos\theta)d\theta[2mv\cos\theta]\cos\theta = 4nmv_R^2 \int_0^{\pi/2} \cos^3\theta\, d\theta$$

$$= \frac{4nmv^2R}{4}\int_0^{\pi/2} (\cos 3\theta + 3\cos\theta)d\theta$$

$$= nmv^2R\left[\frac{\sin 3\theta}{3} + 3\sin\theta\right]\Big|_0^{\pi/2} = \frac{8nmv^2R}{3}$$

7. Ans. 6.00

Sol. $\beta = \frac{D\lambda}{d}$ and $\theta = \frac{D\lambda}{d(D)} = \frac{\lambda}{d}$

$$\frac{\theta_1}{\theta_2} = \left(\frac{\lambda_1}{\lambda_2}\right)\left(\frac{d_2}{d_1}\right)$$

$$\frac{0.04}{\theta_2} = \frac{\lambda}{\left(\frac{\lambda}{\mu}\right)}\left(\frac{(d/2)}{d}\right) \Rightarrow \theta = 0.06$$

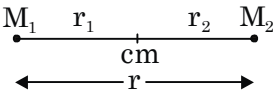
8. Ans. 20.00

Sol. $r_1 = \frac{M_2 r}{M_1 + M_2}$

$$r_2 = \frac{M_1 r}{M_1 + M_2}$$

$$\frac{GM_1M_2}{r^2} = \omega^2 r_2$$

$$\omega = \sqrt{\frac{GM_1}{r^2 r_2}}$$



$$T = \frac{2\pi\sqrt{r^2 r_2}}{\sqrt{GM_1}} = \frac{2\pi\sqrt{r^3}}{\sqrt{G(M_1 + M_2)}}$$

$$r = \sqrt[3]{G(M_1 + M_2)\left(\frac{I}{2\pi}\right)^2}$$

PART–2 : CHEMISTRY SOLUTION

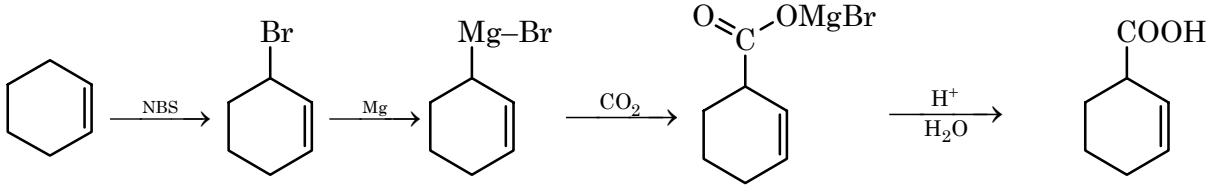
SECTION-I

1. Ans.(C)

Sol. In case of (C) option, CH_3O^- is a strong nucleophile in DMSO and confirms S_N2.

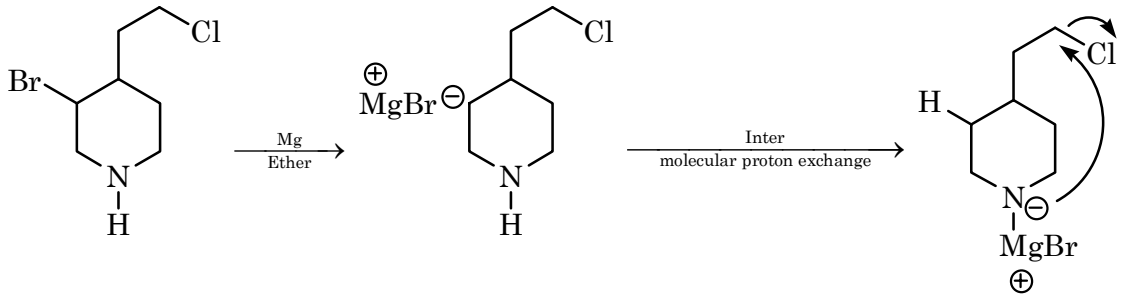
2. Ans.(D)

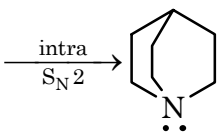
Sol.



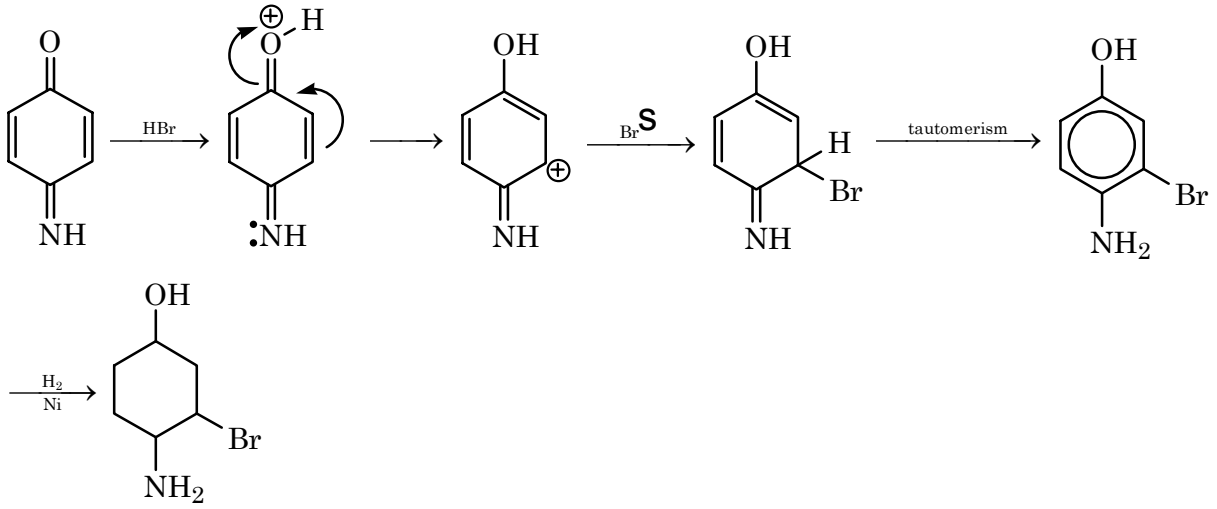
3. Ans.(B)

Sol. (i)

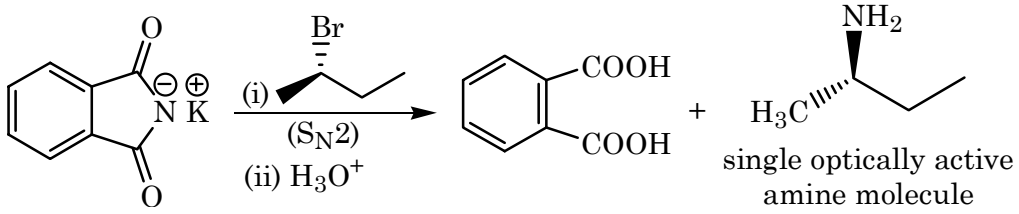


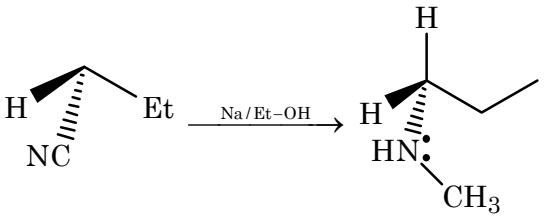


optically inactive

(ii) 

Multiple stereoisomers, optically active of amine as 3 chiral centres are generated.

(iii) 

(iv) 

This is optically inactive 2° amine due to umbrella inversion at N.

4. Ans.(D)

Sol. $\frac{r_X}{r_M} = \frac{0.16a}{0.5a} = 0.32$, so X can occupy only O-Voids (ideal rr is 0.414) but not T-void (ideal rr is 0.225)

and $\frac{r_X}{r_M} = \frac{0.25a}{0.5a} = 0.5$, Y can't occupy either O-voids or T-voids.

On doping X atoms in O-voids, Packing fraction as well as density of crystal will increase as dimensions do not change (no distubing of M-M contact allowed) but total mass increases in the unit cell (we cant comment about X being heavier than M atoms).

5. Ans.(C)

Sol. (A) Initially, mms of OH^- = 100 , mms of H^+ = 10, $[OH^-]$=90/200 => pOH=0.346
Finally, mms of OH^- = 100 , mms of H^+ = 30, $[OH^-]$=70/200 => pOH=0.456
So, change in pH = 0.111 (small change)

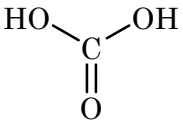
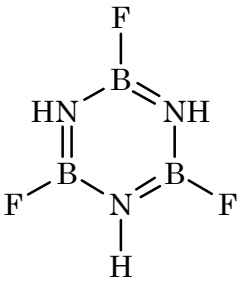
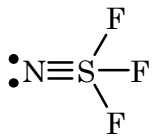
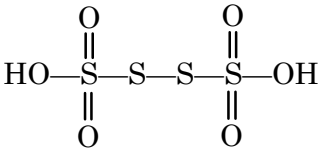
(B) After initial reaction, mms of HS^- = 20 , mms of H_2S = 20 (its a buffer of HS^-/H_2S)
So, change in pH will be very small.

(C) Initially, mms of OH^- = 10 , mms of H^+ = 0 , $[OH^-]$=10/200 $\Rightarrow$ pH=12.7
Finally, mms of OH^- = 10 , mms of H^+ = 20 , $[H^+]$=10/200 $\Rightarrow$ pH=1.3
So, change in pH = 11.4 (very large)

(D) Initially, mms of AcOH = 30, mms of AcO^- = 20 , mms of H_2S = 20 (its a buffer)
So, change in pH will be very small.

6. Ans.(C)

Sol.

Compounds	Structures	A	B	C	D	E
H_2CO_3		×	✓	✓	✓	×
$B_3N_3H_3F_3$		×	✓	✓	✓	×
SNF_3		✓	×	×	×	✓
$H_2S_4O_6$		✓	×	✓	×	✓

7. Ans.(B)

Sol. (i), (ii), (iii), (vii) are the correct combinations

Magnetite	Fe_3O_4
Cryolite	Na_3AlF_6
Copper pyrite	$CuFe_2S_4$
Galena	PbS
Asbestos	$CaMg(SiO_3)_4$
Lunar caustic	$AgNO_3$
Dolomite	$MgCO_3.CaCO_3$

8. Ans.(C)

Sol. Statements (i), (ii), (iv) are correct.

CrO_2Cl_2 gas is liberated (orange red colored), its reaction with NaOH gives yellow solution of Na_2CrO_4 partially covalent halides like $SnCl_4$ do not give this test.

9. Ans.(B)

10. Ans.(A,D)

Sol. (09 to 10)

(i) Osazone D undergoes acidic hydrolysis to create OSONE (first Carbon is aldehydic and second carbon is ketonic) which on further reaction with Zn/CH_3COOH only reduces the aldehydic group to give a ketose.

(ii) As Osazones of glucose, mannose (aldehydic) and fructose (ketonic) are identical, and reduction using Na-Hg produces stereoismers, so among B and C we must obtain Glucose or Mannose, hence, A is fructose.

(iii) & (iv) Using HNO_3 , oxidation of C-1 and C-6 carbons to –COOH group makes Glucaric acid (optically active) and Mannaric Acid (optically active), Hence B is Glucose and C is Mannose

11. Ans.(A)

Sol. $\Delta G = -nEF$

$-827 \times 1000 = -4 \times E \times 96500$ ($\because$ $4e^-$ are exchanged in redox reaction)

$$E = \frac{827000}{4 \times 96500}$$

$E = 2.14$ volts

12. Ans.(A,C)

Sol. (A) EMF = (–0.25) + (–0.34) = negative value

$\Rightarrow$ Non–spontaneous cell

(B) EMF = (–0.34) + (0.8) = positive value

$\Rightarrow$ Spontaneous cell

(C) EMF = (–0.8) + (0) = negative value

$\Rightarrow$ Non–spontaneous cell

(A) EMF = (–0.76) + (0.8) = positive value

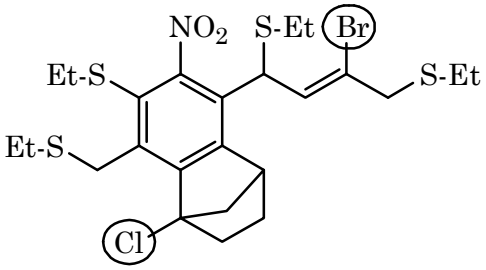
$\Rightarrow$ Spontaneous cell

SECTION-II

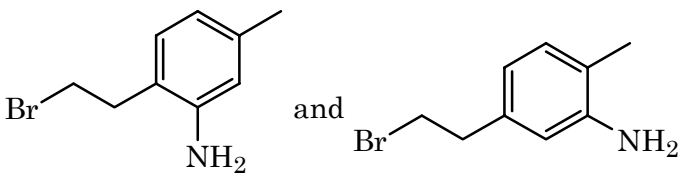
1. Ans.(42)

2. Ans.(4.25 or 5.25)

Sol. S_N2 at Vinylic carbon and at bridge-head carbon of non-fused bicyclo rings NOT POSSIBLE, but S_NAr via carbanion occurs at phenylic 'Cl', Product 'A' of the reaction sequence-1 is,



S_NAr via Benzyne give ipso substitution and cine substitution products (two structural isomers in this case) which on further reaction with HBr/ROOR (Kharasch Effect) do not give any stereoproducts as no new chiral centre is generated, hence final products for reaction seq-2 are



So, $x = \frac{80 \times 1}{35.5 \times 1} = 2.25$ and z = 2, Hence value of (x + z) = 4.25

3. Ans.(0.25)

Sol. Straight line passing through origin have rate $\propto$ [conc]^1 & parallel to X–axis have rate $\propto$ [conc]^0

Hence

r_1 = Rate = $K_1[A]^1 \Rightarrow X = 1$

r_2 = Rate = $K_2[B]^0 \Rightarrow Y = 0$

r_2 = Rate = $K_3[C]^1 \Rightarrow Z = 1$

4. Ans.(2.50)

Sol. $P_T = P_A^\circ X_A + P_B^\circ X_B$ (According to Raoult's law)

$450 = 300\,(X) + 700\,(1 - X)$

$X = 0.625$

Now Y is the mole fraction of A in vapour mixture

$$\frac{Y}{1-Y} = \frac{X_A}{X_B}.\frac{P_A^\circ}{P_B^\circ} = \frac{0.625}{0.375}.\frac{300}{700} \Rightarrow Y = 0.417$$

$$P = \frac{450+300}{2} = 375 \text{ mm of Hg} \qquad (\because \text{ Similar triangles})$$

$$\frac{P.Y}{100X} = \frac{375 \times 0.417}{100 \times 0.625} = 2.5$$

5. Ans.(2.00)

Sol. At constant volume $\begin{cases} Z = \dfrac{N}{V}.\sigma.\sqrt{2}\,u_{avg} \\ \lambda = \dfrac{k_B}{\sqrt{2}\sigma}\left(\dfrac{T}{P}\right) \end{cases}$

N = number of gas molecules

σ = collision area (depends on molecular size only)

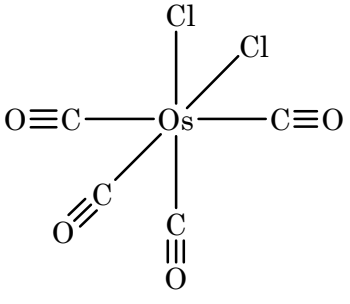
u_{avg} = average velocity of gas molecules = $\sqrt{\dfrac{8RT}{\pi M}}$

6. Ans.(1.14)

Sol. x = 1 $\Rightarrow$ No possibility of geometrical isomerism in 1st complex.

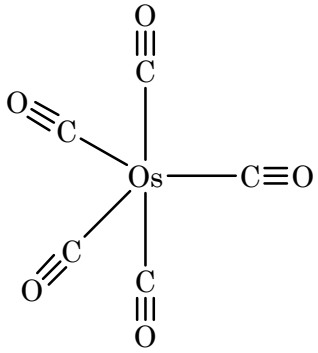
x = 2 $\Rightarrow$ Possible with octahedral & TBP geometries.

x = 3 $\Rightarrow$ Geometry of 1st complex will not have 90° & 180° angles.



$\Rightarrow$ cis–isomer octahedral, P = 8 atoms

$OsCl_2(CO)_4$



$\Rightarrow$ TBP geometry, Q = 7 atoms in equitorial plane.

$Os(CO)_5$

7. Ans.(0.20)

Sol. x = 1 (i.e. $CaCO_3$)

y = 5 (all except carbonate of Potassium easily undergo decompostion)

z = 6 (all liberate CO_2 with mineral acid)

8. Ans.(3.47 to 3.48)

Sol. PCl_5 and PBr_5 both exist in ionic form in solid state, both give H_3PO_4 and HX on hydrolysis. Bromine is liberated from NaBr on reaction with conc H_2SO_4 (this is not possible with NaCl). Hence, a = 5, b = 3, c = 4 and atomic weight of Bromine is 80 amu. Hence 80/23 value is 3.48

PART-3 : MATHEMATICS SOLUTION

SECTION-I

1. Ans. (C)

2. Ans. (B)

Sol. Let $t=\frac{1}{(x+y)^2}$

$$\frac{dt}{dx}=-\frac{2}{(x+y)^3}\cdot\left(1+\frac{dy}{dx}\right)$$

so $$\frac{1}{(x+y)^3}\frac{dy}{dx}+\frac{x}{(x+y)^2}-x^3=\frac{-1}{(x+y)^3}$$

$$\Rightarrow \quad -\frac{1}{2}\frac{dt}{dx}=-tx+x^3$$

$$\frac{dt}{dx}=2tx-2x^3$$

$$\text{I.F.} = e^{\int -2x\,dx} = e^{-x^2}$$

$$\Rightarrow \quad \frac{1}{(x+y)^2}=x^2+1+ce^{x^2}$$

3. Ans. (C)

Sol. Let $P(\lambda, \lambda-2, \lambda)$ and $a\left(\mu-2, \frac{\mu}{2}, \frac{\mu}{2}\right)$

Then $$\frac{\lambda-\mu+2}{2}=\frac{\lambda-2-\frac{\mu}{2}}{1}=\frac{\lambda-\frac{\mu}{2}}{2}$$

$$\Rightarrow \quad \lambda-\mu+2=2\lambda-\mu-4$$

$$\lambda-\mu+2=\lambda-\frac{\mu}{2}$$

$$\Rightarrow \quad \mu=4,\ \lambda=6$$

P(6, 4, 6) Q(2, 2, 2)

$PQ = \sqrt{16+4+16} = 6$

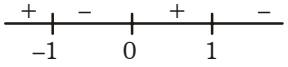
4. Ans. (C)

Sol. $g(x)=f(2-x^2)+f(x^2)$

$g'(x)=f'(2-x^2)(-2x)+f'(x^2).2x$

$= (2x)(f'(x^2)-f'(2-x^2))$

Since $f''(x) < 0 \Rightarrow f'(x)$ is a decreasing function



Hence g(x) is increasing in $(-\infty, -1) \cup (0, 1)$

5. Ans. (D)

Sol. Required Probability

$$=\frac{P(HTHT)+P(THTH)}{1-\left(P(HHHH)+P(TTTT)\right)}$$

$$=\frac{\left(\frac{2}{3}.\frac{1}{3}.\frac{2}{3}.\frac{1}{3}\right)2}{1-\left(\left(\frac{2}{3}\right)^4+\left(\frac{1}{3}\right)^4\right)}=\frac{1}{8}$$

6. Ans. (C)

Sol. Let $x^4 = t$

$t^2 - (k-1)t + 5 = 0$

for max number of real root both the roots of equation must be positive

$D > 0 \Rightarrow (k-1)^2 - 4(5) > 0 \Rightarrow \quad k > 1 + \sqrt{20}$

also $k > 1$

Least possible integral value is 6

7. Ans. (D)

Sol. Slope forms : $y = mx \pm \sqrt{16m^2 - 9}$

$y = mx \pm \sqrt{-9m^2 + 16}$

Comparing $\Rightarrow m = 1, m = -1$

$\therefore$ 4 common tangents : 2 of slope 1, 2 of slope -1

$\therefore$ Sum of squares = 4

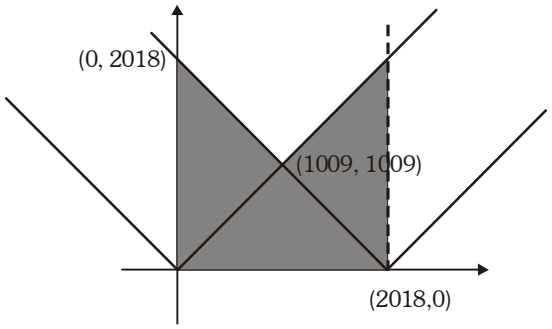
8. Ans. (C)

Sol. Total number of rectangles is equal to no. of ways of selecting 2 diagonals out of 8 diagonals (which pass through center)

${}^8C_2 = 28$

Solution for Q. No. 9 & 10

Sol.



$\int_0^{2018} f(x)\,dx$ = shaded area

$= 3\left(\frac{1}{2}(2018)(1009)\right) = 3\,(1009)^2$

similarly $\int_0^{2018} g(x)\,dx = 2018 \int_0^1 g(x)\,dx$

$= \frac{2018}{4} = \frac{1009}{2}$

9. Ans. (D)

10. Ans. (D)

Paragraph-2 for Q.no. (11 & 12)

$1's \to 4, \quad 2's \to 3, \quad 0's \to 2$

So possible elements for diagonal are

$$\begin{bmatrix} 2 & 1 & 1 \\ - & 2 & 0 \\ - & - & 2 \end{bmatrix} \to 3$$

$$\begin{bmatrix} 2 & 1 & 1 \\ - & 0 & 2 \\ - & - & 0 \end{bmatrix} \to 9$$

$$\begin{bmatrix} 2 & 2 & 1 \\ - & 1 & 0 \\ - & - & 1 \end{bmatrix} \to 18$$

sum of traces of all matrix

$= 6(3) + 2(9) + 4(18) = 108$

11. Ans. (C)

12. Ans. (C)

SECTION - II

1. Ans. 1.00

2. Ans. 2.47

Sol. Given curves are symmetric about $y = x$

$f'(x) = 2x = 1$

$x = \frac{1}{2}, \; y = 2 + \frac{1}{4} = \frac{9}{4}$

Shortest distance of f(x) from $y = x$ is

$$\frac{\left|\frac{9}{4} - \frac{2}{4}\right|}{\sqrt{2}} = \frac{7}{4\sqrt{2}}$$

Hence S. D = $\frac{7}{2\sqrt{2}} = \frac{7\sqrt{2}}{4}$

3. Ans. 3.00

Sol. $x = 1$; $y = 1, 2, 3$ (possibilities)

$\therefore$ Required probability = $\frac{1}{3} \times 1 = \frac{1}{3}$

4. Ans. 7.00

Sol. E = 2 balls drawn, none of them is white.

E_1 = both balls red

E_2 = both balls black

E_3 = one red, one black

$$P(E_1/E) = \frac{P(E/E_1).P(E_1)}{P(E/E_1).P(E_1) + P(E/E_2).P(E_2) + P(E/E_3).P(E_3)}$$

$$= \frac{\frac{{}^3C_2}{{}^7C_2}}{\frac{{}^3C_2}{{}^7C_2} + \frac{{}^2C_2}{{}^7C_2} + \frac{{}^3C_2.{}^2C_1}{{}^7C_2}} = \frac{{}^3C_2}{{}^3C_2 + {}^2C_2 + {}^3C_1.{}^2C_1}$$

$$= \frac{3}{3 + 1 + 3.2} = \frac{3}{10}$$

5. Ans. 4.00

Sol. $(a-1)x + 0y + z = \alpha$ ... (1)

$x + (b-1)y + 0z = \beta$ ... (2)

$0x + y + (c-1)z = \gamma$ ... (3)

For no unique solution D = 0

$$\begin{vmatrix} (a-1) & 0 & 1 \\ 1 & (b-1) & 0 \\ 0 & 1 & (c-1) \end{vmatrix} = 0$$

$(a-1)(b-1)(c-1) + 1 = 0$

$\therefore a = 2; b = 2; c = 0$

Hence, $|a + b + c| = 4$

6. Ans. 6.00

Sol. $$6\sum_{r=1}^{10}\frac{24r^2+2}{64r^6-48r^4+12r^2-1}$$

$$= 6\sum_{r=1}^{\infty}\frac{24r^2+2}{(4r^2-1)^3} = 6\sum T_r$$

$$\Rightarrow \quad T_r = \frac{24r^2+2}{(2r-1)^3(2r+1)^3}$$

$$= \frac{1}{(2r-1)^3} - \frac{1}{(2r+1)^3}$$

$$\sum_{r=1}^{\infty} T_r = 1$$

7. Ans. 2.70

Sol. Triangle will not be formed if

$L_1 \,||\, L_2 \Rightarrow 2 = -\frac{1}{k} \Rightarrow k = -\frac{1}{2}$

or $L_1 \,||\, L_3 \Rightarrow 2 = -\frac{k}{2} \Rightarrow k = -4$

or $L_2 \,||\, L_3 \Rightarrow \frac{-1}{k} = \frac{-k}{2} \Rightarrow k^2 = 2 \Rightarrow k = \pm\sqrt{2}$

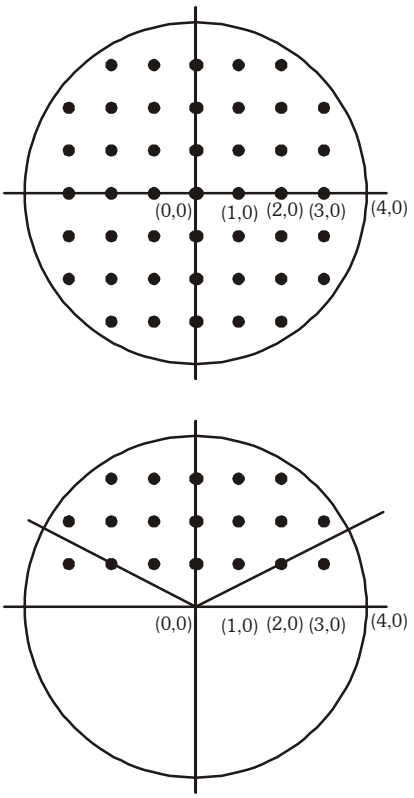
or L_1, L_2, L_3 are concurrent

$\Rightarrow \quad k = -\frac{6}{5}, 3$

$\text{sum} = -\frac{1}{2} - 4 - \frac{6}{5} + 3 = -2.7$

8. Ans. 3.00

Sol. By ploting A = 45 and B = 15



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
07-05-2021

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE [SCORE-II]

PAPER-2

PART-1 : PHYSICS — SOLUTION

SECTION-I

1. Ans. (D)

Sol. We know that frequency of orbital motion

$$f \propto \frac{1}{n^3} \text{ and given } f_1 = \frac{1}{27} f_2$$

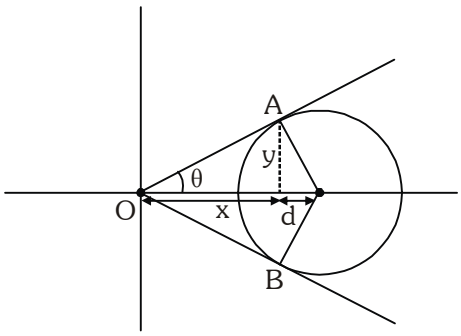
$$\Rightarrow \left(\frac{n_2}{n_1}^3\right) = \frac{f_1}{f_2} \Rightarrow \frac{n_2}{n_1} = \left(\frac{1}{27}\right)^{1/3} = \frac{1}{3}$$

2. Ans. (B,C,D)

Sol. $\sin\alpha = \frac{\ell}{4y_0^2}$, $\alpha = \sin^{-1}\left(\frac{\ell}{4y_0^2}\right)$

3. Ans. (A,B,C)

Sol. $\sin\theta = \frac{3}{5} \Rightarrow \theta = 37°$



$$d = 3\cos 53° = \frac{9}{5}$$

$$y = 3 \sin 53° = \frac{12}{5}$$

At position A, frequency is maximum & at position B frequency is minimum

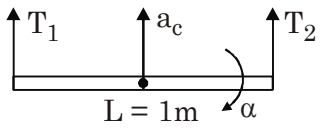
4. Ans. (A, B)

Sol. $a_C + \frac{L}{2}\alpha = a_1$

$$a_C - \frac{L}{2}\alpha = a_2$$

$$\Rightarrow \alpha = \frac{(a_1 - a_2)}{L} = 2 \text{rad/s}^2$$

$a_1 = 3$ m/s^2 $\quad$ $a_2 = 1$ m/s^2



$$\alpha = \frac{(a_1 + a_2)}{2} = 2\,\text{m/s}^2$$

$$T_1 + T_2 - mg = ma_c$$

$$(T_1 - T_2)\frac{L}{2} = \frac{mL^2}{12}\alpha$$

$\Rightarrow T_1 = 185/6$ N and $T_2 = 175/6$ N

5. Ans. (B,C)

Sol. $MS(2\theta - T) = M2S(T - \theta)$

$$2\theta - T = 2T - 2\theta$$

$$T = \frac{4\theta}{3}$$

and $M_A S\Delta T = M_B 2S\Delta T \Rightarrow \frac{M_A}{M_B} = 2$

6. Ans. (C)

Sol. $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

% error

$$= \frac{\Delta R_1}{R_1} \times 100 + \frac{\Delta R_2}{R_2} \times 100 + \frac{\Delta(R_1 + R_2)}{(R_1 + R_2)} \times 100$$

$$= 1 + 2 + \frac{.03 + 0.12}{9} \times 100$$

$= 3 + 1.67 = 4.67\%$.

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7. Ans. (C,D)

8. Ans. (C)

Sol. $F = v_e \frac{dm}{dt} = \rho A v^2$

$$\tau = \frac{(\rho A v^2)H}{2}$$

9. Ans. (B)

Sol. Lenz law.
Direction of induced current is such that it oppose increase or decrease the magnetic field in any direction across loop.

10. Ans. (A)

Sol. If any charge is present outside, then charge on outer surface will be distributed non-uniformly. But if there is no charge present outside, then charge on outer surface will be distributed uniformly. This is irrespective of location of the charge inside.
If charge is at center, then charge on inner surface will be distributed uniformly, but if charge is displaced from center, then charge on inner surface will be distributed non-uniformly. This is irrespective of whether the charge is present outside or not.

11. Ans. (B)

Sol. $K.E. = \frac{1}{2} m\omega^2 (A^2 - x^2)$

$$P.E. = \frac{1}{2} m\omega^2 x^2$$

12. Ans. (C)

Sol. From graph, we see that graph 2 is decreasing faster than graph 1. Hence $\lambda_1 > \lambda_2$.

SECTION-III

1. Ans. 6

Sol. $i = \sqrt{5}A$

$$\frac{q_m^2}{2C} = \frac{q^2}{2C} + \frac{1}{2} L i^2 \Rightarrow q_{max} = 6\ C$$

2. Ans. 1

Sol. One particle t/2 before max height, other t/2 after max height
Relative velocity perpendicular to line joining them = gt
Relative separation = ut
Relative angular velocity = gt/ut = g/u

3. Ans. 5

Sol. Area = πab
$= \pi\ 10 \times 10^3 \times 10 \times 10^{-3} = 100\ \pi$

4. Ans. 2

Sol. $2\sqrt{2g\ell \sin\theta} = \sqrt{2g\ell}$

$4 \sin\theta = 1$

$8 \sin\theta = 2$

5. Ans. 3

Sol. $h\nu = 17$ eV
Now for metal,
$h\nu = \phi + (KE)_{max}$
$\Rightarrow$ 17 eV = ϕ + 2 eV $\Rightarrow \phi = 15$ eV.

6. Ans. 6

Sol. Capillary height $h = \frac{2T\cos\theta}{r\rho g}$

$\Rightarrow h = 59.6$ mm

here 59.6 mm is greater than the protruding part hence water will rise in the capillary of insufficient height 25 mm.

Now, $R = \frac{2T}{hrg} = 0.6$ mm

7. Ans. 1

Sol. $M = I\frac{\pi R^2}{2}\sqrt{2}$ and $2\pi R = L$

Hence $M = \frac{IL^2\sqrt{2}}{8\pi}$

8. Ans. 5

Sol. Beat frequency

$$= 2\left(\frac{\omega_1 - \omega_2}{2\pi}\right) = 2\left(\frac{10\pi}{2\pi}\right) = 2k$$

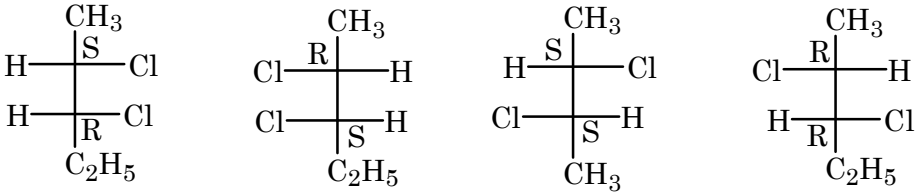
Hence k = 5

PART–2 : CHEMISTRY SOLUTION

SECTION-I

1. Ans.(A,B,D)

Sol.



(I) (II) (III) (IV)

2. Ans.(A,B,C)

Sol. Acid–base reaction.

3. Ans.(A,B,C,D)

Sol. Theory based.

4. Ans.(A,B,D)

Sol. (B) Brownian movement is shown by colloidal solution due to collision between colloidal particle themselves and with that of particles of dispersion medium. Tyndall effect is due to scattering of light by colloidal particles.

(D) is correct because, higher the gold number, less will be protective power.

(C) is not correct because when liquid is dispersed in liquid, it is called emulsion.

5. Ans.(A,C)

Sol. On dilution the molarity of the solution decreases and hence molar conductivity increases. So choice (A) is correct.

(C) On dilution, ionic mobility increases due to decreases in interionic forces of attraction, hence (C) is correct.

(D) But the conducting power of ions per cubic centimeter decreases, hence specific conductance decreases as volume of solution increases.

Hence, choice (D) is incorrect.

6. Ans.(A,B,D)

Sol. $CaCO_3(s) \overset{K_{p_1}}{\rightleftharpoons} CaO(s) + CO_2(g)$

$CO_2(g) + C(s) \overset{K_{p_2}}{\rightleftharpoons} 2CO(g)$

$K_{p_1} = p_{CO_2} = 4\,atm \qquad K_{p_2} = \dfrac{(p_{CO})^2}{p_{CO_2}}$

$p_{CO} = \sqrt{K_{p_2} p_{CO_2}} = \sqrt{4 \times 2 \times 10^{-2}}$

$p_{CO} = 0.28\ atm$

$\therefore$ (A) and (B) are correct answers.

7. Ans.(A,C,D)

Sol. $Al(CH_3)_3$ and B_2H_6 has three centre two electron bond.

$(AlCl_3)_2$ has three centred $4e^-$ bond.

8. Ans.(B,C,D)

Sol. N–F bond is less polar than N–H bond.

9. Ans.(A)

Sol. Theory based.

10. Ans.(C)

Sol. Theory based.

11. Ans.(A)

Sol. (P) Carbocation intermediate is formed but there is rearrangement.

(Q) Carbocation intermediate is formed but there is no rearrangement.

12. Ans.(D)

Sol. (P) $H_2O(s) \rightleftharpoons H_2O(\ell)$ 1 atm and 273 K.

$\because$ Endohtermic reaction $\therefore q > 0$

also volume decrease $\therefore w > 0$

$\therefore \Delta S_{sys} > 0$ $\Delta U > 0$

Also since reaction is at equilibirum $\therefore \Delta G = 0$

(Q) $\because$ isolated condition $\therefore q = 0$, alsoexpansion against vacuum $\therefore w = 0$ $\therefore \Delta U = 0$, also I.G.

$\therefore$ Temperature remains constant.

Also, entropy increases

Also for physical process

$dG = VdP - SdT$

$\therefore$ it will be < 0

(R) $\because$ isolated container $\therefore q = 0$ and $w = 0$

$\because$ Gases are gettting mixed $\therefore$ entropy of system is increasing

Also $\Delta U = 0$ ($\because \Delta U = q + w$)

Also $\because$ mixing is spontaneous $\therefore \Delta G < 0$

(S) $\because$ Reversible heating and cooling

$\therefore$ initial a find state will be same.

hence atll state functions will remain same.

Also q and w for heating and cooling will be opposite of each other

$\therefore$ overall $q = 0$

$\therefore$ overall $w = 0$

SECTION-III

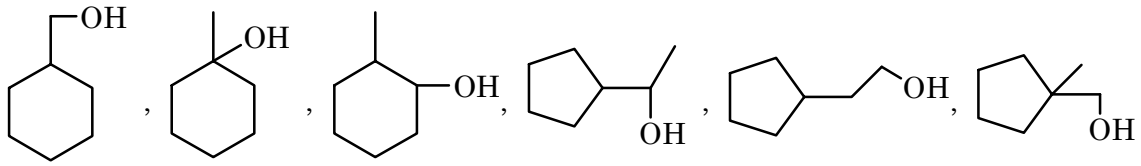
1. Ans.(9)

Sol. Theory based.

2. Ans.(8 or 9)

Sol. Theory based.

3. Ans.(6)

Sol. 

4. Ans.(3)

Sol. $\underset{\text{Black}}{\underset{\text{'A'}}{Cu(OH)_2.CuCO_3}} + 2H_2SO_4 \longrightarrow \underset{\text{'C'}}{CuSO_4} + \underset{\text{'B'}}{CO_2}$ $Cu^{2+} \longrightarrow 3d^9$

$H_2S + \underset{\text{'C'}}{CuSO_4} \longrightarrow CuS + H_2SO_4$

$\underset{\text{'D'}}{CuS} + 2HNO_3(dil.) \longrightarrow Cu(NO_3)_2 + H_2S$

$Cu(NO_3)_2 + 2NH_4OH \longrightarrow Cu(OH)_2 + 2NH_4NO_3$

$\underset{\text{Soluble}}{Cu(NO_3)_2} + 4NH_4OH \longrightarrow \underset{\text{Soluble (F)}}{[Cu(NH_3)_4]}(NO_3)_2 + 2H_2O$

$2Cu^{2+} + K_4[Fe(CN)_6 \longrightarrow \underset{\text{'E'}}{Cu_2[Fe(CN)_6]} + 4K^+$

5. Ans.(4)

Sol. Graphite, black phosphorus, rhombic sulphur, white tin and Cl_2 are most stable state but $\Delta_f H^o \neq 0$ for black phosphorus.

6. Ans.(4)

Sol. $2KMnO_4 + 5H_2O_2 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5O_2 + 8H_2O$

$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 10CO_2 + 8H_2O$

10 mole of $H_2C_2O_4$

$\therefore$ 4 moles of $KMnO_4$

$\frac{100}{1000} \times M = 4$

$M = 40$

7. Ans.(6)

Sol. A buffer solution should contain weak acid and its conjugate base or weak base and its conjugate acid.

8. Ans. (5)

Sol. $[K^+] = \frac{M_1V_1 \times x + M_2V_2}{x \times V_1 + V_2}$

$0.06 = \frac{0.1 \times 5 \times x + 0.02 \times 25}{x \times 5 + 25}$

$x = 5$

PART-3 : MATHEMATICS SOLUTION

SECTION–I

1. Ans. (A)

Sol. $f(2-x) = f(2+x)$

$x \to 2-x$

$f(x) = f(4-x)$ .....(i)

differentiate w.r.t. x

$f'(x) = -f'(4-x)$ .....(ii)

Put x = 1/2 in (ii)

$f'\left(\frac{1}{2}\right) = f'\left(\frac{7}{2}\right) = f'(1) = f'(2) = f'(3) = 0$

also

Put x = 1, 2

Now consider f'(x) = y.

by Roll's theorem we can say in $\left[\frac{1}{2},1\right],[1,2],\left[2,\frac{7}{2}\right],\left[\frac{7}{2},3\right]$

f''(x) = 0 has minimum 4 roots.

2. Ans. (A,B,D)

Sol. $f(x) = |(x-2)(x-3)|$

$$= \begin{cases} x^2-5x+6, & 0 \le x \le 2 \\ -x^2+5x-6; & 2 < x \le 2.4 \end{cases}$$

$$f'(x) = \begin{cases} 2x-5, & 0 \le x \le 2 \\ -2x+5, & 2 < x \le 2.4 \end{cases}$$

$f'(x) = 0 \Rightarrow x = \frac{5}{2} \notin [0,2.4]$

at x = 2 m = 0

at x = 0, M = 6.

3. Ans. (A,C)

Sol. $\int_0^{\pi} (8g(x)\sin^2 x - (g(x))^2)dx = 6\pi$

$\Rightarrow \int_0^{\pi} 16\sin^4 x dx - \int_0^{\pi} (g(x) - 4\sin^2 x)^2 dx = 6\pi$

$\Rightarrow 32\left(\frac{3\pi}{16}\right)\int_0^{\pi} (g(x) - 4\sin^2 x)^2 dx = 6\pi$

$6\pi - \int_0^{\pi} (g(x) - 4\sin^2 x)^2 dx = 6\pi$

$\Rightarrow g(x) = 4\sin^2 x$

M = 4.

4. Ans. (A,B)

Sol. $T_{r+1} = {}^{10}C_r \, 2^{\frac{(10-r)}{2}} . 3^{\frac{r}{5}}$

this is rational, if 10 – r is even

r is a multiple of 5.

r = 5 $\Rightarrow \frac{10-r}{2}$ is not an integer.

r = 10 $\Rightarrow$ 10 – r = 0 and $\frac{r}{5} = 2$

r = 0 $\Rightarrow \frac{10-r}{2} = 5$

Therefore the sum of the rational term is

$\Rightarrow {}^{10}C_0 \, 2^5 + {}^{10}C_{10} \, 3^2 = 32 + 9 = 41$

5. Ans. (A,C,D)

Sol. $z^2 - \bar{z}^2 = 4i$

$\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2i}\right) = 1$ ....(1)

if z = x + iy

then (1) is xy = 1

the circle $x^2 + y^2 = 4$ intersect the rectangular hyperbola in four points which are symmetrical about origin.

6. Ans.(A,C)

Sol. Probability (0H or 2H or 4H)

$Q = P^4 + 6P^2(1-P)^2 + (1-P)^4$

$= 8P^4 - 16P^3 + 12P^2 - 4P + 1$

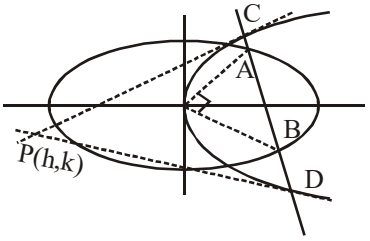
$Q = \frac{(2P-1)^4 + 1}{2}$ now check from options.

7. Ans. (D)

Sol. ky = 2a (x + h) (chord of contact from P to parabola)

homogenize with ellipse

$\frac{x^2}{4a^2} + \frac{y^2}{a^2} - \left(\frac{ky - 2ax}{2ah}\right)^2 = 0$



coefficient of x^2 + coefficient of $y^2 = 0$

and replace h by x and k by y

$\Rightarrow 5x^2 - y^2 = 4a^2$

8. Ans. (B,C)

Sol. $\pi - \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) + \sin^{-1}\left(\frac{2x}{1+x^2}\right) - \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$= \frac{2\pi}{3}$

$\Rightarrow \cos^{-1}(\cos 2\theta) + \tan^{-1}(\tan 2\theta) - \sin^{-1}(\sin 2\theta) = \frac{\pi}{3}$

$\theta \in \left(0, \frac{\pi}{4}\right) \Rightarrow 2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6} \Rightarrow x = \frac{1}{\sqrt{3}}$

$\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \Rightarrow \theta = \frac{7\pi}{18} \Rightarrow x = \tan\left(\frac{7\pi}{18}\right) = \tan 70°$

$\theta \in \left(-\frac{\pi}{4}, 0\right) \Rightarrow \theta = -\frac{\pi}{6} \Rightarrow x = \left(\frac{-1}{\sqrt{3}}\right)$

no. of solution is 3.

SECTION–I

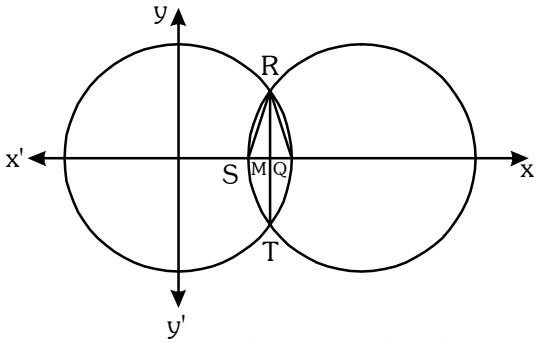
9. Ans. (C)

Sol. (P) $y - y^2 = a^2$

$D > 0 \Rightarrow a \in \left(-\frac{1}{2}, \frac{1}{2}\right)$

$|\alpha - \beta| = 1$

(Q)



Variable circle $(x-1)^2 + (y-0)^2 = r^2$.

$x^2 + y^2 - 2x + (1 - r^2) = 0;\ 0 < r < 2.$

$R : \left\{\frac{2-r^2}{2}, \frac{r\sqrt{4-r^2}}{2}\right\}$

$\Delta QRS = \frac{1}{2}\frac{r^2\sqrt{4-r^2}}{2}$

maximum when $r = \sqrt{\frac{8}{3}}$

(R) $AB = 2r\sqrt{\frac{S_1}{S_1 + r^2}} = 2.3\sqrt{\frac{9}{9+9}}$

$= 3\sqrt{2} \Rightarrow T = 2$

(S) $L_T = \sqrt{d^2 - (r_1 + r_2)^2} = 24$

10. Ans. (D)

Sol. (P) $\bar{a} = \left(3\hat{i} - 4\hat{j} + 2\hat{k}\right)$

$\bar{b} = -4\hat{i} - 7\hat{j} + 4\hat{k}$

projection of $\bar{a}$ on $\bar{b}$

$= \frac{\bar{a}.\bar{b}}{|b|} = \frac{-12 + 28 + 8}{\sqrt{81}} = \frac{24}{\sqrt{81}} = \frac{8}{3}$

(Q) $\bar{\alpha} + \bar{\beta} + \bar{\gamma} = 0$

$\bar{\alpha} + \bar{\beta} = -\bar{\gamma}$

$\Rightarrow \left(\bar{\alpha}\right)^2 + \left(\bar{\beta}\right)^2 + 2\bar{\alpha}.\bar{\beta} = \left(\bar{\gamma}\right)^2$

$\Rightarrow 2.3.5\cos\theta = 49 - 9 - 25 = 15$

$\cos\theta = \frac{1}{2}.$

(R) $\left(\bar{A} + \bar{B}\right) \times \left(\bar{A} - \bar{B}\right) = \bar{A} \times \bar{A} - \bar{A} \times \bar{B} + \bar{B} \times \bar{A} - \bar{B} \times \bar{B}$

$= -2\left(\bar{A} \times \bar{B}\right)$

$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 5\hat{i} - 7\hat{j} + 11\hat{k}$

$\left|\bar{A} \times \bar{B}\right| = \sqrt{195}$

$\left|\left(\bar{A} \times \bar{B}\right) \times \left(\bar{A} - \bar{B}\right)\right| = 2\sqrt{195}$

(S) $(\bar{a} + \bar{b} + \bar{c})^2 = \bar{a}^2 + \bar{b}^2 + \bar{c}^2 + 2(\bar{a}.\bar{b} + \bar{b}.\bar{c} + \bar{c}.\bar{a})$

$0 = 3 + 2\ (\bar{a}.\bar{b} + \bar{b}.\bar{c} + \bar{c}.\bar{a})$

$\Rightarrow \frac{-3}{2}$

11. Ans. (B)

Sol. (P) $xy(ydx + xdy) + x^2y^2(2ydx - xdy) = 0$

$\frac{1}{x^2y^2}d(xy) + \frac{2dx}{x} - \frac{dy}{y} = 0$

$2\ \ell n\ |x| = \ell n\ |y| + \frac{1}{xy} + C$

(Q) $ydx + x\,dy + xy(y\,dx - xdy) + x^2y^2\,(ydx + xdy) = 0$

$\Rightarrow (1 + x^2y^2)d(xy) + xy(ydx - xdy) = 0$

$\Rightarrow \left(\frac{1}{x^2y^2} + 1\right)d(xy) + \frac{dx}{x} - \frac{dy}{y} = 0$

$\Rightarrow \frac{-1}{xy} + xy + \ln|x| - \ln|y| + c$

(R) $(x^2 + y^2 + 2x)dx + 2y\,dy = 0$

$(x^2 + y^2)\,dx + d(x^2 + y^2) = 0$

$dx + d(\ln(x^2 + y^2)) = 0$

$x + \ln(x^2 + y^2) = C$

(S) $y\,dx - x\,dy = xy(x\,dy + y\,dx)$

$\frac{dx}{x} - \frac{dy}{y} = d(xy)$

$d(\ell n(x)) - d(\ell n(y)) = d(xy)$

$\ell n\left|\frac{x}{y}\right| = xy + c.$

12. Ans. (B)

Sol. (P) $f'(x) + f(x) = x$ fi $x \ge f(x)$

$f'(x) - f(x) = -x$ if $x \le f(x)$

$$\Rightarrow f(x) = \begin{cases} x - 1 + e^{-x}; & x \le 0 \\ x + 1 - e^{-x} & x \ge 0 \end{cases}$$

point of inflection (0, 0)

$\alpha + \beta = 0$

(Q) $\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$

(R) for $a \in \left(\frac{-37}{4}, -3\right) \cup (-3, 3) \cup \left(3, \frac{37}{4}\right)$

f(x) = 0 has 4 real roots.

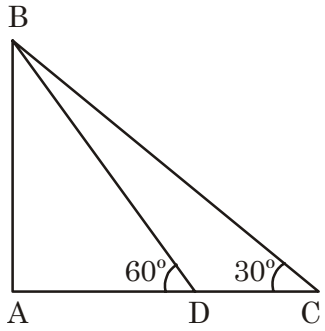
(S) for $a \in \left\{\frac{-37}{4}, -3, \frac{37}{4}, 3\right\}$

for 1 = 0 hes 3 real roots.

SECTION–III

1. Ans. 7

Sol. ΔABD



$\tan 60^\circ = \frac{AB}{AD} = \sqrt{3}$, let AB = x metre

$\Rightarrow \quad AD = \frac{x}{\sqrt{3}}$

ΔACB

$$\tan 30^\circ = \frac{x}{AC} = \frac{1}{\sqrt{3}}$$

$\Rightarrow \quad AC = \sqrt{3}\,x$

Now $CD = AC - AD$

$$CD = \sqrt{3}\,x - \frac{x}{\sqrt{3}} = \frac{2x}{\sqrt{3}} \text{ metre}$$

given that car covered $\frac{2x}{\sqrt{3}}$ metre in 15 minutes.

So it will cover $AD = \frac{x}{\sqrt{3}}$ in 7.5 minutes P = 7.5 minutes

[P] = 7 minutes

2. Ans. 4

Sol. Let r is radius of circle

$x^2 + (y - 3)^2 = r^2$

$$e^2 = 1 - \frac{9}{16} = \frac{7}{16}$$

$$e = \frac{\sqrt{7}}{4} \Rightarrow ae = \sqrt{7}$$

foci are $\left(\pm\sqrt{7}, 0\right)$ since circle passes through $\left(\sqrt{7}, 0\right)$

$7 + 9 = r^2 \Rightarrow \quad r = 4$

3. Ans. 2

Sol. $A_1 = \frac{a+b}{2}$, $G_1 = a.\left(\frac{b}{a}\right)^{\frac{1}{3}}$, $G_2 = a.\left(\frac{b}{a}\right)^{\frac{2}{3}}$

$G_1^3 = a^2b, G_2^3 = b^2a$

$G_1G_2 = a^2\left(\frac{b}{a}\right) = ab$

$$\Rightarrow \frac{G_1^{\,3} + G_2^3}{G_1G_2A_1} = \frac{ab(a+b).2}{ab.(a+b)} = 2.$$

4. Ans. 1

Sol. $x - x^2 + x[x] = x - x(x - [x])$

$= x(1 - \{x\})$

$x = \left(2 + \sqrt{3}\right)^n = {}^nC_0n^{2n} + {}^nC_12^{n-1}\sqrt{3} + ...,$

let $x_1 = \left(2 - \sqrt{3}\right)^n$

$= {}^nC_02^n - {}^nC_12^{n-1}\sqrt{3} + {}^nC_22^{n-2}\left(\sqrt{3}\right)^2 + ...$

$x + x_1 = 2\left({}^nC_02^n - {}^nC_22^{n-2}\left(\sqrt{3}\right)^2 + ...\right)$

= Even integer

$x_1 \in (0,1) \forall n \in N$

$\Rightarrow [x] + \{x\} + x_1$ = Even integer

$\Rightarrow \{x\} + x_1$ = Integer.

Since, $\{x\} \in (0, 1)$, $x_1 \in (0, 1)$

$\Rightarrow \{x\} + x_1 \in (0, 2) \Rightarrow \{x\} + x_1 = 1$

$x_1 = 1 - \{x\}$

$\Rightarrow x(1 - \{x\}) = x.x_1$

$= \left(2+\sqrt{3}\right)^n \left(2-\sqrt{3}\right)^n = 1.$

5. Ans. 7

Sol. $A = (I + B)(I - B)^{-1}$

$AA^T = (I + B)(I - B)^{-1}((I - B)^T)^{-1}(I + B)^T$

$= (I + B)(I - B)^{-1}(I + B)^{-1}(I - B)$

$= (I + B)((I + B)(I - B))^{-1}(I - B)$

$= (I + B)((I - B)(I + B))^{-1} (I - B)$

($\because$ I + B, I – B are commutative)

$\therefore AA^T = (I + B)(I + B)^{-1}(I - B)^{-1}(I - B) = I$

$\therefore AA^T = I$

$|AA^T| = |I| = 1 \Rightarrow |A|^2 = 1$

as $|A| > 0 \Rightarrow |A| = 1$

$\det(2A) - \det(\text{adj } (A)) = 8 \det(A) - \det(A)^2$

$= 8(1) - (1)^2 = 7.$

6. Ans. 7

Sol. $\int \frac{4}{(3-2x)^2}\left(\frac{3+2x}{3-2x}\right)^{\frac{-1}{3}} dx$

let $\frac{3+2x}{3-2x} = z^3$

$\frac{4}{(3-2x)^2} dx = z^2 dz$

$I = \int z^2 z^{-1} dz = \frac{z^2}{2} + c$

$= \frac{3}{6}\left(\frac{3+2x}{3-2x}\right)^{\frac{2}{3}} + c \Rightarrow \alpha = 6, \beta = 2, \gamma = 3$

$\alpha + \gamma + \beta = 9 - 2 = 7$

7. Ans. 1

Sol. $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ell n \left(\frac{n^2 + (n-1)^2}{n^2 + k^2}\right)^k$

$\lim_{n\to\infty} \frac{1}{n} \ell n \left(\frac{n^2}{n^2+1^2} \cdot \frac{(n^2+1^2)^2}{(n^2+2^2)^2} \cdot \frac{(n^2+2^2)^3}{(n^2+3^2)^3} \cdots \left(\frac{n^2+(n-1)^2}{n^2+n^2}\right)^n\right)$

$\lim_{n\to\infty} \frac{1}{n} \left(\ln \left(\frac{n^2(n^2+1^2)(n^2+2^2).....(n^2+(n-1)^2)}{(2n^2)^n}\right)\right)$

$\lim_{n\to\infty} \frac{1}{n} \left[\ell n \frac{1}{2}\left(\frac{1}{2}+\frac{1^2}{2n^2}\right) \cdot \left(\frac{1}{2}+\frac{2^2}{2n^2}\right) \cdot \left(\frac{1}{2}+\frac{(n-1)^2}{2n^2}\right)\right]$

$\lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n-1} \ell n \left(\frac{1}{2} + \frac{r^2}{2n^2}\right)$

$= \int_0^1 \underbrace{1.\ell n \left(\frac{1+x^2}{2}\right)}_{I} dx$

$S = -2\left[x - \tan^{-1} x\right]_0^1 = -2\left[1 - \frac{\pi}{4}\right]$

$L = \frac{\pi}{2} - 2$

$|[L]| = 1.$

8. Ans. (5)

Sol. $6x + 8yy' = 0, y' = \frac{-3x}{4y}$

$4(yy'' + (y')^2) = -3$

differentiating again and substitute y'

$3 + 4(y')^2 + 4yy'' = 0$

hence $3 + \frac{9x^2}{4y^2} + 4yy'' = 0$

$3y^2 + \frac{9x^2}{4} + 4y^3 y'' = 0$

$\frac{3}{16}(3x^2 + 4y^2) + y^3 y'' = 0$

$3x^2 + 4y^2 = 12$

$y^3 y'' = \frac{-9}{4}$ at every point on ellipse.

$\left|y^3 y''\right| = \frac{9}{4} = \frac{P}{Q}$

$\Rightarrow P - Q = 5$